Laser Fault Injection in Memristor-Based Accelerators for AI/ML and Neuromorphic Computing

Muhammad Faheemur Rahman, Wayne Burleson
Electrical and Computer Engineering, University of Massachusetts Amherst
Email: faheemur@umass.edu, burleson@umass.edu

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Background & Motivation:

Memristive crossbar arrays (MCAs) have become promising hardware platforms for energy-efficient in-memory and neuromorphic computing due to their ability to perform analog matrix–vector multiplication.

However, their non-volatile analog nature introduces new security and reliability challenges. Physical fault injection, particularly Laser Fault Injection (LFI), can manipulate or reveal stored weights by generating photocurrents within the silicon substrate.

This work investigates how LFI perturbs analog computation and stored weights in MCA-based accelerators and examines their accuracy, performance, and robustness under such attacks.

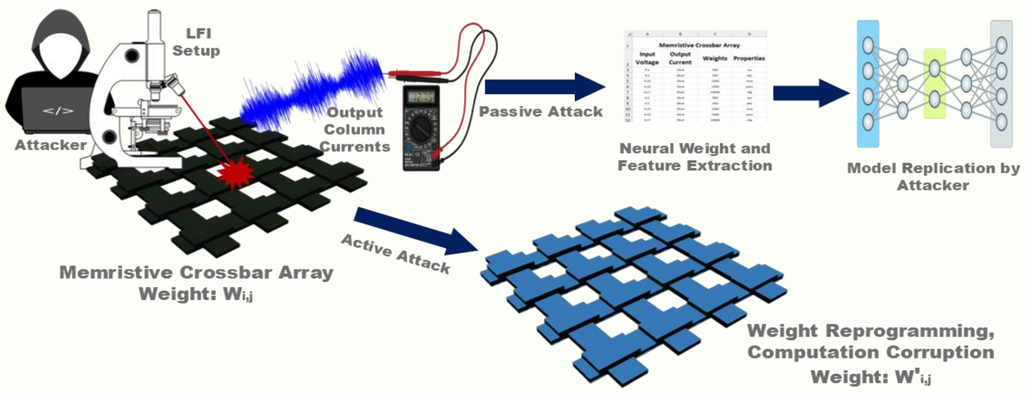

Fig. 1: LFI Adversarial Attack Flow on Memristive Crossbar Array

Research Objectives:

- Develop a realistic threat model for laser-induced perturbations in memristive arrays.
- Model laser-induced photocurrents and their effect on memristor conductance.
- Quantify information leakage and permanent weight shifts due to LFI across different current injection levels.
- Evaluate the performance, accuracy, and robustness of MCAs under different fault magnitudes.
- Propose directions for fault-resilient and secure analog hardware design, emphasizing sensor integration and adaptive protection schemes.

Methodology:

Simulation Platform: HSPICE using 45 nm CMOS process node.

Array Configuration: 256×128 1T1R MCA (memristor + NMOS selector).

Fault Injection Model:

- LFI modeled as localized current injection (10 μA–1.2 mA) with adjustable spot size (1–50 μm).
- Injected currents emulate photon-induced electron–hole generation near active regions.

Memristor Models:

- **Linear resistor** model (5–20 kΩ) for initial current estimation.
- **TEAM** (Threshold Adaptive Memristor) model for nonlinear, realistic behavior under current-controlled faults.

Analysis:

- Record column output currents before and after injection.
- Apply linear regression to estimate internal conductances representing neural weights.
- Study resistance (or equivalently, conductance) shifts for weight modification at high injection currents.

Key Findings:

Weight Extraction:

- Linear regression achieved ≤ 0.35% error when testing within 10–40 μA.
- Faults in high-resistance cells yield larger ΔI shifts, improving precision.

Weight Modification:

- TEAM model shows that currents > 50 μA noticeably alter conductances.
- At 1.2 mA, resistance increases ~140%, changing stored weights.

Performance & Robustness:

- Low-level faults primarily expose weights without major degradation.
- High-magnitude faults cause accuracy loss and performance drift, indicating limited fault tolerance in current MCA architectures.

Inj. Current (μA)	ΔI (μA) at 5 kΩ	ΔI (μA) at 10 kΩ	ΔI (μA) at 12 kΩ	ΔI (μA) at 15 kΩ	ΔI (μA) at 20 kΩ
10	2.29	1.28	1.09	0.89	0.68
15	3.43	1.93	1.64	1.34	1.03
20	4.59	2.58	2.19	1.79	1.37
30	6.91	3.88	3.31	2.70	2.07
40	9.25	5.21	4.43	3.62	2.78
⋮	⋮	⋮	⋮	⋮	⋮

Table 1. Weight–Fault Current Relationship across different Memristor Resistance

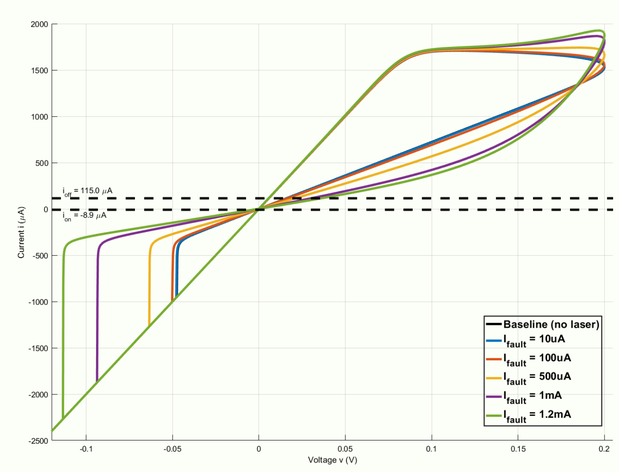

Fig. 2: Memristor I-V: Fault Current Sweep

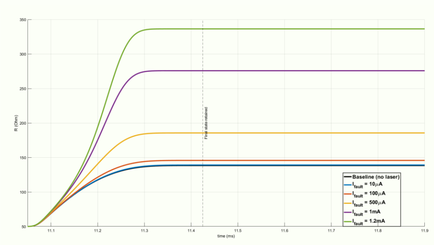

Fig. 3: Resistance R(x) under Laser Fault

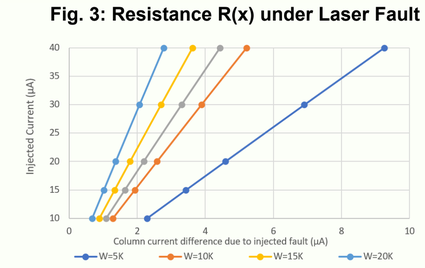

Fig. 4: Relation between Weights & Injected Fault Current

Countermeasure Strategies Against LFI:

1. Fault Detection (On-Chip Sensors)
   - Detects voltage drop (IR-drop) or clock freeze caused by laser hits.
   - VDD, clock and delay sensors flag abnormal timing or waveform shifts.
   - Photodiodes and Thermal sensors detect laser light or overheating.
2. Protection Response
   - Alarm controller triggers counteractions when faults are detected.
   - Write-protection circuits (WE lock, fuse, or PUF-authenticated) block unauthorized writes.
   - Threshold control limits voltage & pulse width to stop unwanted updates.

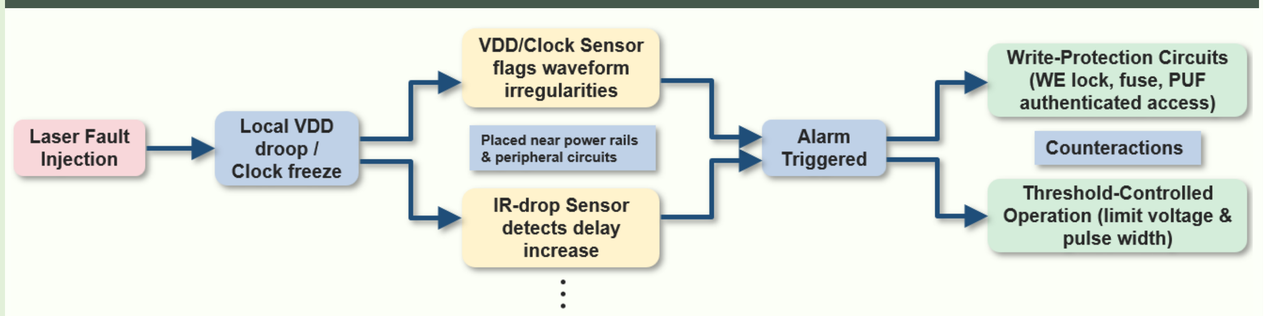

Fig. 5: System-Level Detection and Protection Flow Against Laser Fault Injection

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Reference

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Conclusions:

- LFI can both extract and reprogram analog weights in MCAs.
- μA-level photocurrents lead to model extraction attacks.
- mA-level injections cause permanent weight alterations.
- Findings reveal a trade-off between analog efficiency and fault robustness.
- Future work: integrate on-chip sensors, shielding layers, and current limiters for proactive fault detection.